Shaft Component Generator (Version: 2025.2 (Build 292293000, 293))

5/28/2025

Project Info (iProperties)

Calculation

Material

Material		AISI 1050
Modulus of Elasticity	E	210000 MPa
Modulus of Rigidity	G	81000 MPa
Density	ρ	7850 kg/m^3

Calculation Properties

Include			
Yes	Density	ρ	7850 kg/m^3
Yes	Shear Displacement Ratio	β	1.188 ul
	Number of Divisions		1000 ul
	Mode of reduced stress		HMH

Loads

Index	Location	Radial Force				Bending Moment				Continuous Load					Axial Force	Torque	Deflection				Deflection Angle
		Y	X	Size	Direction	Y	X	Size	Direction	Y	X	Size	Direction	Length			Y	X	Size	Direction	
1	122.75 mm	3726.330 N	10238.000 N	10895.053 N	70.00 deg												-3.749 microm	-15.542 microm	15.987 microm	256.44 deg	0.01 deg
2	288.75 mm	-1520.950 N	4178.780 N	4446.964 N	110.00 deg												-1.388 microm	-9.968 microm	10.064 microm	262.07 deg	0.01 deg

Supports

Index	Type	Location	Reaction Force					Yielding	Type	Deflection				Deflection Angle
			Y	X	Size	Direction	Axial Force			Y	X	Size	Direction	
1	Free	30 mm	2427.348 N	8109.843 N	8465.316 N	73.34 deg			User	-0.000 microm	-0.000 microm	0.000 microm	180.00 deg	0.01 deg
2	Fixed	352 mm	-75.205 N	6306.937 N	6307.386 N	90.68 deg			User	-0.000 microm	-0.000 microm	0.000 microm	188.15 deg	0.01 deg

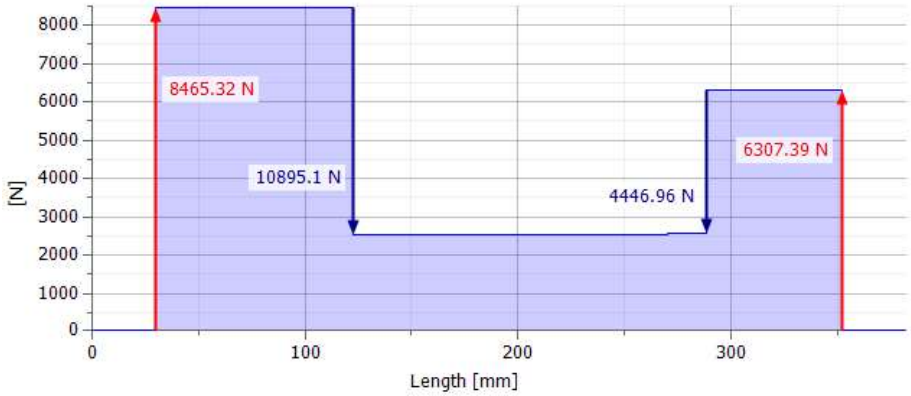
Results

Length	L	382.000 mm
Mass	Mass	14.966 kg
Maximal Bending Stress	σ_B	15.605 MPa
Maximal Shear Stress	τ_S	1.915 MPa
Maximal Torsional Stress	τ	0.000 MPa
Maximal Tension Stress	σ_T	0.000 MPa
Maximal Reduced Stress	σ_{red}	15.874 MPa
Maximal Deflection	f_{max}	17.923 microm
Angle of Twist	φ	0.00 deg

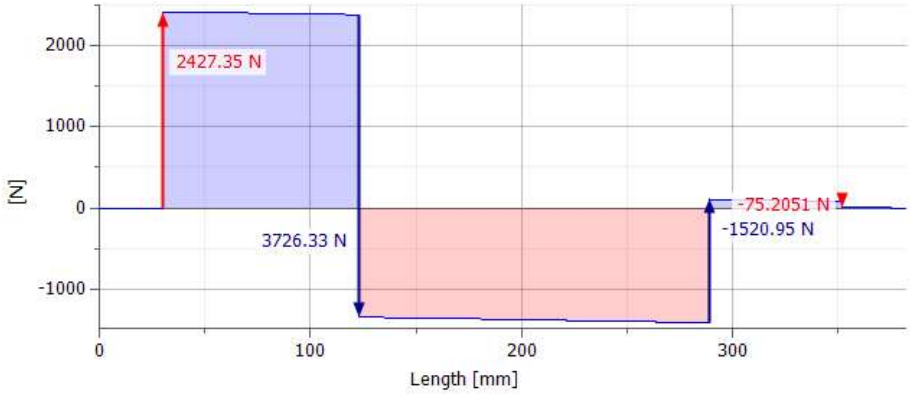
Preview



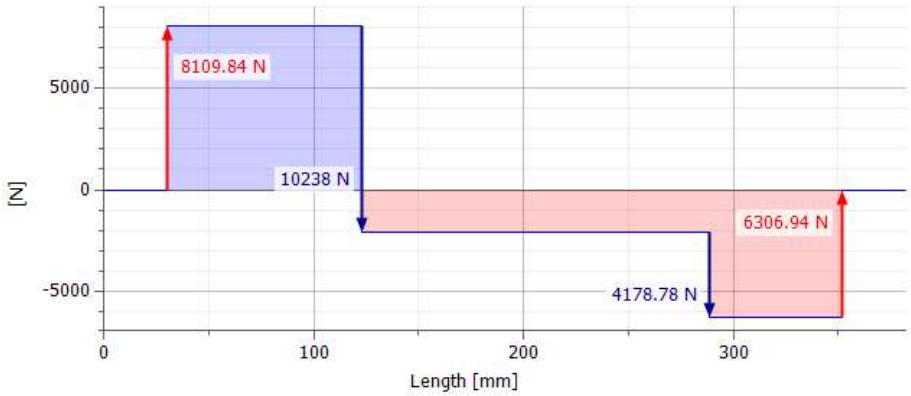
Shear Force



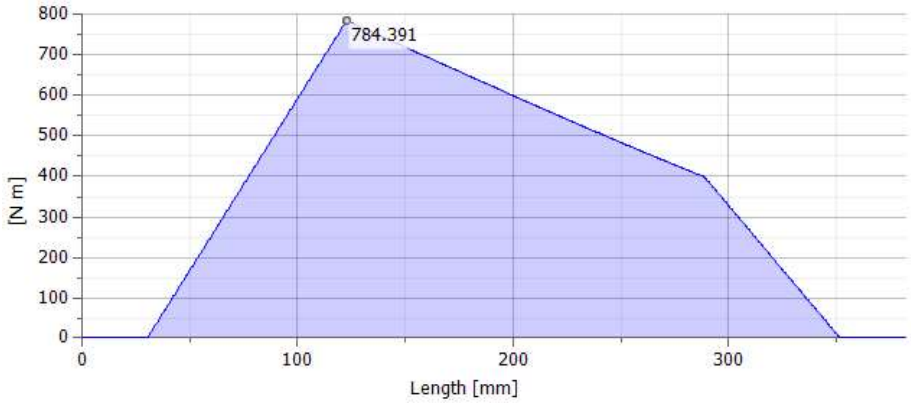
Shear Force, YZ Plane



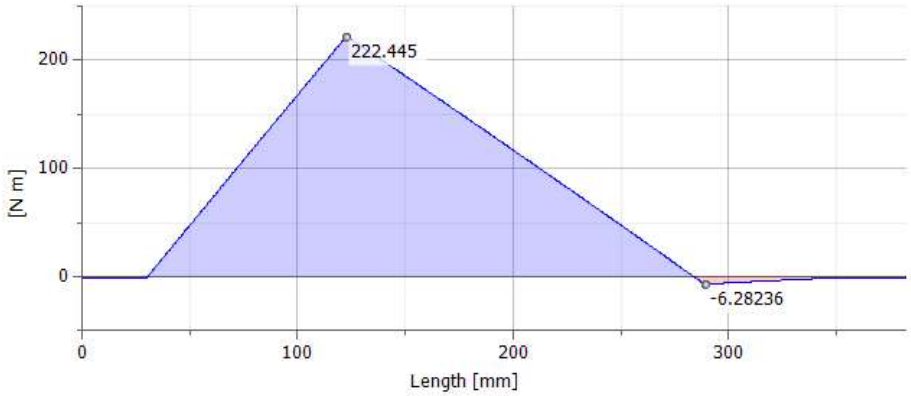
Shear Force, XZ Plane



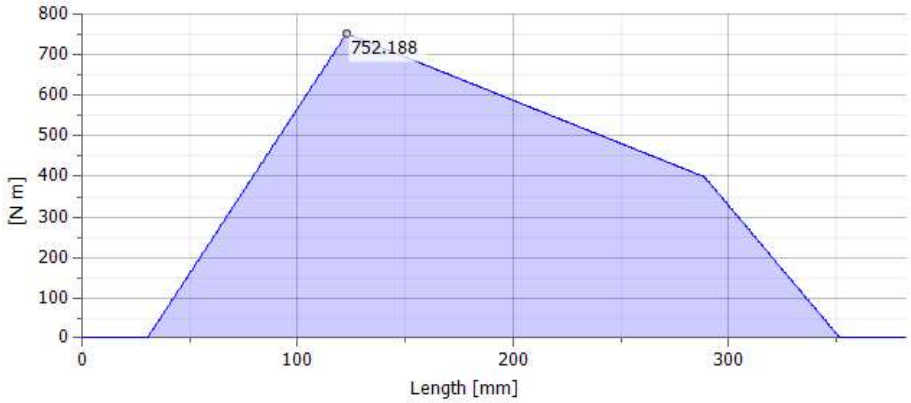
Bending Moment



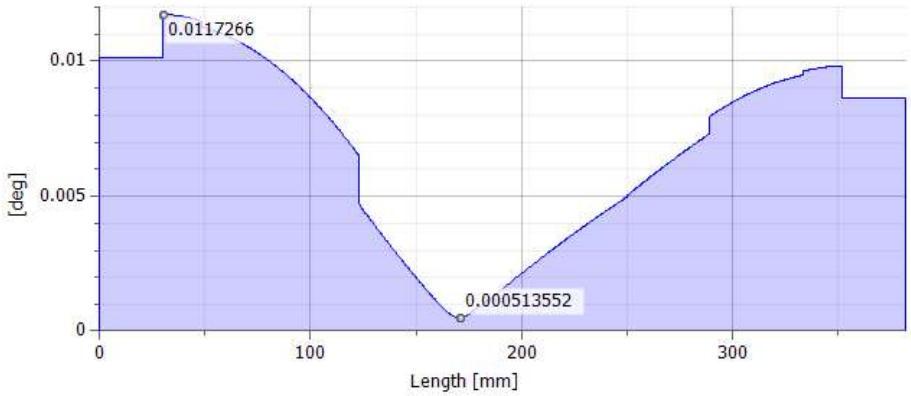
Bending Moment, YZ Plane



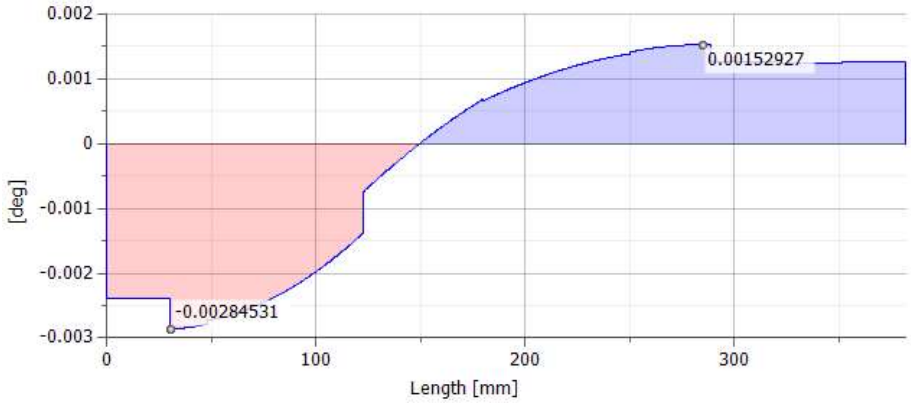
Bending Moment, XZ Plane



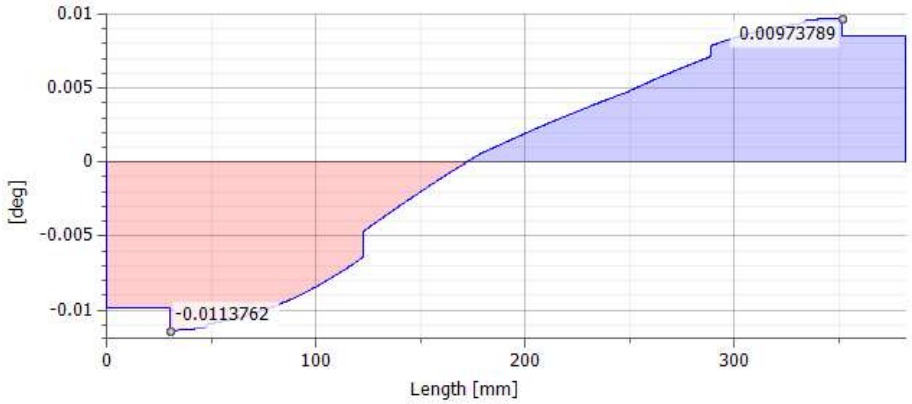
Deflection Angle



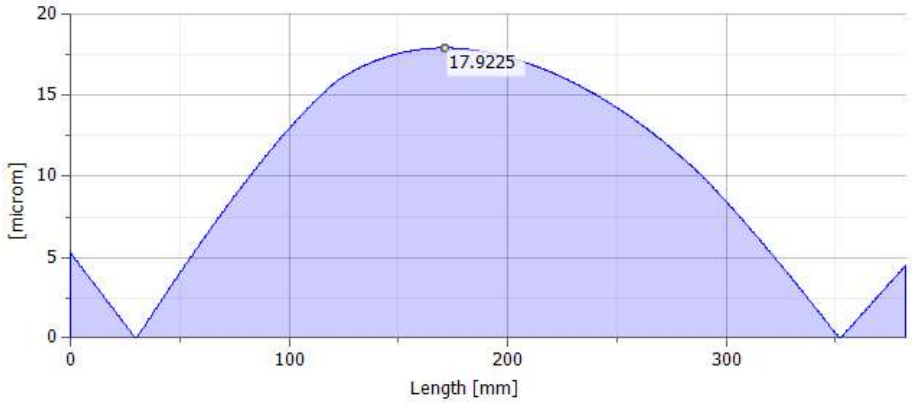
Deflection Angle, YZ Plane



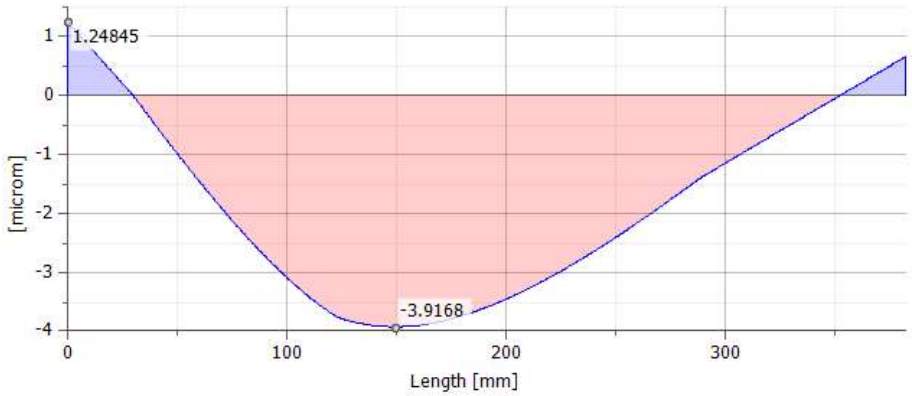
Deflection Angle, XZ Plane



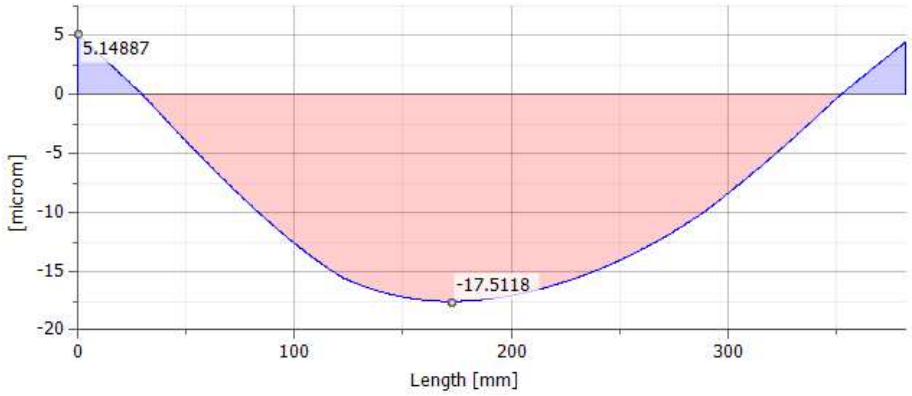
Deflection



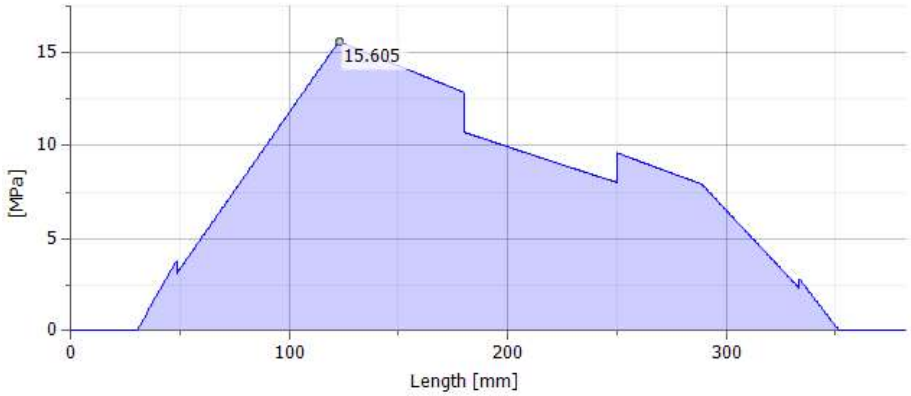
Deflection, YZ Plane



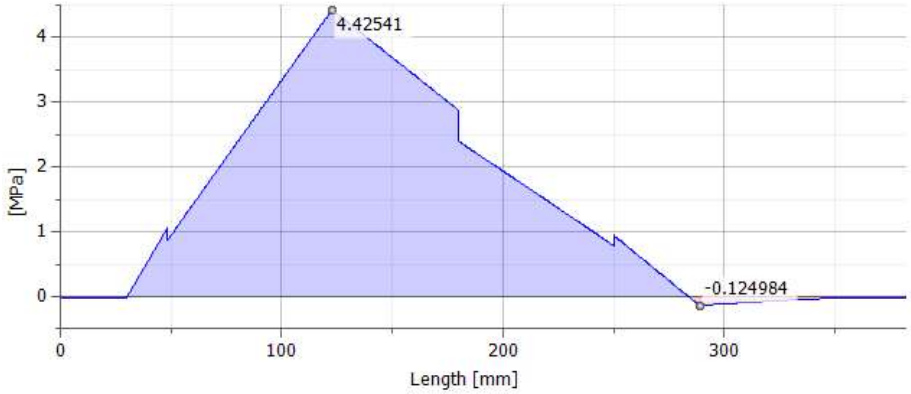
Deflection, XZ Plane



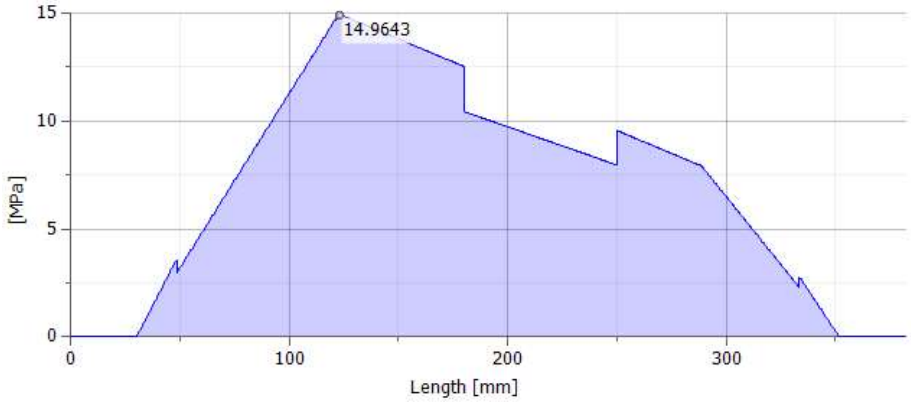
Bending Stress



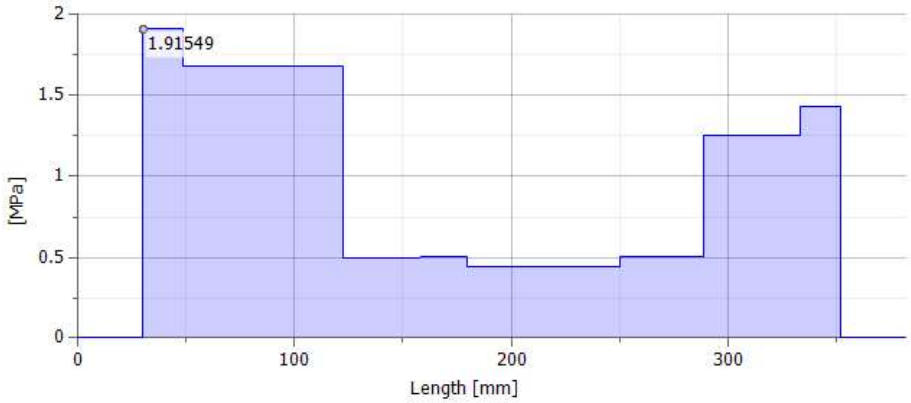
Bending Stress, YZ Plane



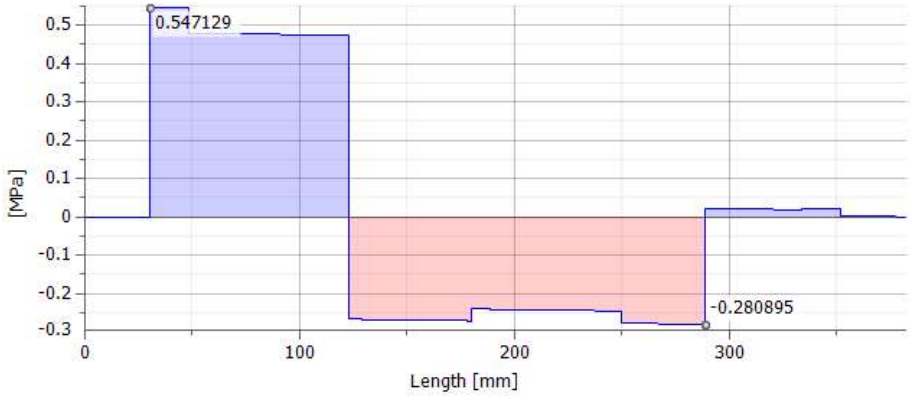
Bending Stress, XZ Plane



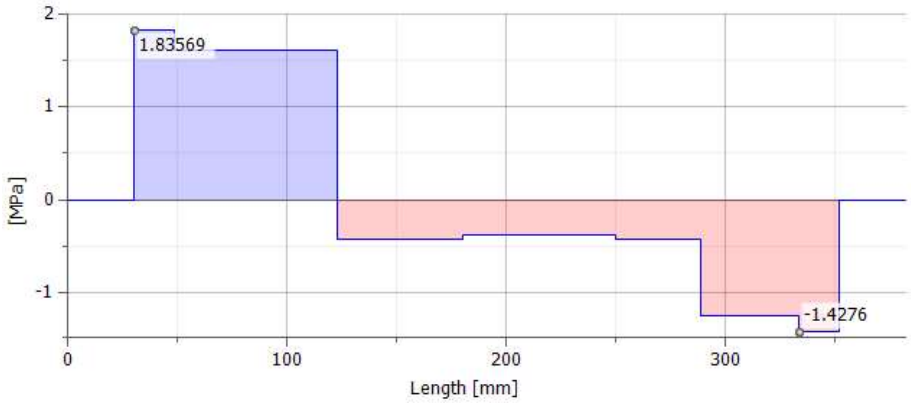
Shear Stress



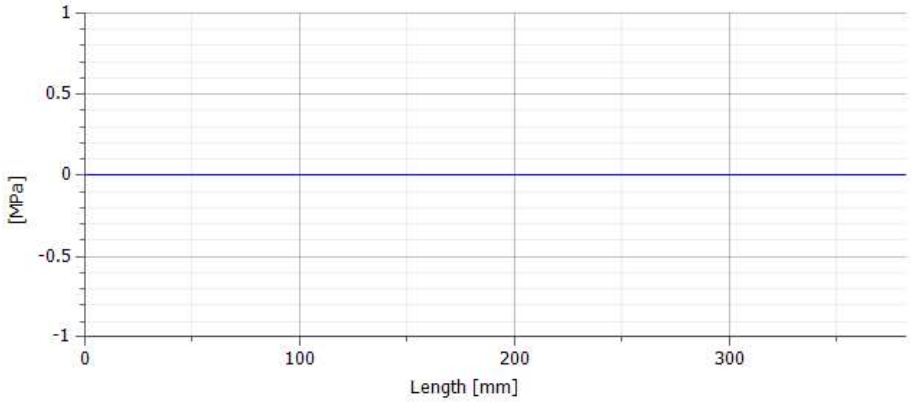
Shear Stress, YZ Plane



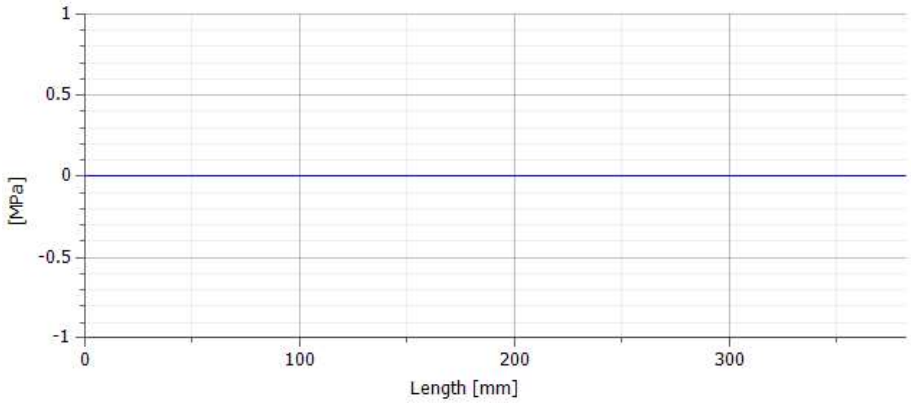
Shear Stress, XZ Plane



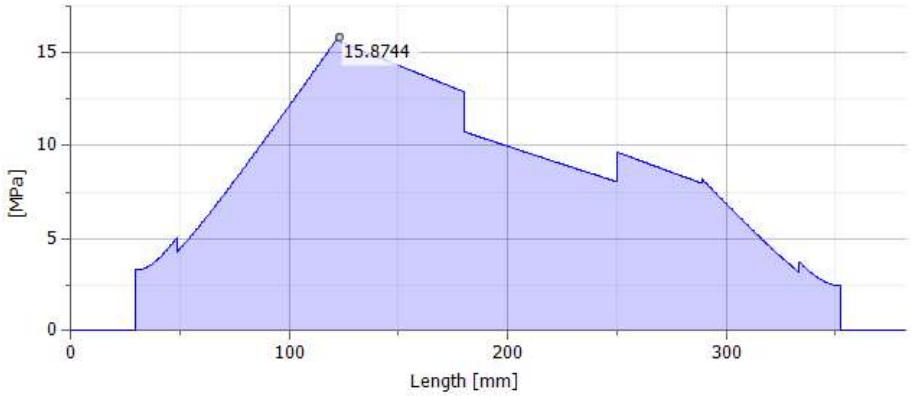
Torsional Stress



Tension Stress



Reduced Stress



Ideal Diameter



Summary of Messages

6:59:12 AM Calculation: Calculated.